University of Zurich[UZH]

ETH

Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Multi-Currency Portfolio Optimization with General Risk Measures

Master Thesis

Anej Rozman

Supervisor: Prof. Patrick Cheridito
Co-Supervisor: Dr. Urban Ulrych

Department of Mathematics, ETH Zürich

Acknowledgments

Abstract

Contents

Chapter 1

Introduction

The median of a dataset is a fundamental measure of central tendency that arises in numerous computational and statistical contexts. In an array of n elements, indexed from 1 to n, the median can be intuitively understood as the "middle" element in the sorted order if n is odd, or the average of the two middle elements if n is even. While this definition naturally suggests sorting the entire array, doing so in $\mathcal{O}(n \log(n))$ time is often unnecessary when the goal is to identify only the median.

Motivation and Previous Work

Several well-known algorithms exist for selecting the median or, more generally, the k-th smallest element of an array. A straightforward method is to sort the array and then pick the median in constant time. However, this approach has a runtime of $\mathcal{O}(n \log(n))$, which can be excessive for large n. More advanced methods, such as the *median-of-medians* algorithm introduced by Blum, Floyd, Pratt, Rivest, and Tarjan [1], achieve a worst-case runtime of $\mathcal{O}(n)$ using a deterministic divide-and-conquer strategy. While optimal in the worst case, the median-of-medians algorithm involves significant overhead, making it less practical in many scenarios. Alternatively, randomized algorithms like Quickselect often achieve linear-time performance on average but can degrade to $\mathcal{O}(n^2)$ in the worst case without additional precautions.

Our Contribution

In this thesis, we present a *randomized* algorithm that computes the median in $\mathcal{O}(n)$ time *with high probability*, thereby combining practical efficiency with strong theoretical guarantees. The key idea is to leverage a carefully chosen random sample of the input array:

1. We extract a subset of elements (the "sample") whose size grows sublinearly with n.
2. Using this sample, we identify pivot elements that, with high probability, closely approximate the true median's position.
3. We then reduce the problem to sorting only a small subset of the original array, thereby achieving near-linear time complexity.

Central to our analysis is the use of **Chebyshev's inequality**. While often introduced in purely statistical contexts, Chebyshev's inequality provides a useful probabilistic bound for ensuring that the sampled elements approximate the median accurately. In essence, it tells us that $\mathbb{P}[|X - \mathbb{E}[X]| \geq k]$ can be kept small if $\mathrm{Var}[X]$ is finite, thereby giving us high confidence that our pivot-based "zoom-in" on the median is correct.

More concretely, we prove the following theorem with our algorithm.

Theorem 1.1 *There exists a randomized algorithm to compute the median that runs in time $\mathcal{O}(n)$ and succeeds with probability $1 - \mathcal{O}\left(n^{-1/4}\right)$.*

Thesis Outline

This thesis is structured as follows:

- **Chapter 3 (Notation and Preliminaries):** We define our notational conventions and offer a brief refresher on Chebyshev's inequality, setting the stage for the randomized analysis to come.
- **Chapter 4 (Algorithm and Analysis):** We detail the randomized median-finding algorithm, present its pseudocode, and provide a rigorous proof of correctness. We also analyze the runtime and show why it remains $\mathcal{O}(n)$ with high probability.
- **?? (Conclusion):** We summarize our findings, discuss the algorithm's theoretical and practical implications, and highlight possible directions for future research.

Chapter 2

Literature Review

In this chapter, we review the relevant literature on multi-currency portfolio optimization. We discuss the contributions and methodologies in the field, highlighting how our work builds upon and extends existing research.

The core of our analysis builds on two recent studies that apply regularization techniques to international asset allocation. Burkhardt and Ulrych (2023) introduce a joint optimization approach within a mean-variance framework. By using L1 and L2 regularization, they show that jointly optimizing assets and currencies leads to better out-of-sample performance than the standard industry practice of using a separate currency overlay.

Extending this framework, Lucescu and Ulrych (2026) consider an Expected Shortfall optimization framework to better manage downside tail risk. To fix the numerical instability common in Expected Shortfall minimization, they apply similar regularization methods. Their findings reveal a dimensional trade-off. Joint optimization works best in small asset universes, but the separate currency overlay approach becomes more effective for larger universes due to its built-in dimensionality reduction.

Reproducing and comparing these two baseline frameworks serves as the starting point for the extensions proposed in this thesis.

Chapter 3

Notation and Preliminaries

Definitions and Notation

Arrays and Median

Throughout this thesis, we denote the input array of size n by

$$a = (a[1], a[2], \dots, a[n]).$$

We may also use $a[1 \dots n]$ to denote an array a of length n. Our goal is to find the *median* of a. Formally, we define:

- If n is odd, the median is the element in the $\lceil n/2 \rceil$-th position of the array when sorted in non-decreasing order.
- If n is even, the median is the element in the $\lceil n/2 \rceil$-th position of the sorted array.

Random Variables and Expectation

In the analysis of our randomized median-finding algorithm, we shall frequently refer to *random variables*. A random variable X is a function from the sample space Ω (representing all possible outcomes of a probabilistic experiment) to the real numbers $\mathbb{R}$.

Expectation. The *expected value*, or *mean*, of X is defined by

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega)\, \mathbb{P}[\omega],$$

in the discrete case. If X has a probability density function f_X (in the continuous case), then

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x\, f_X(x)\, dx.$$

Intuitively, $\mathbb{E}[X]$ indicates the "average" or "central" value of the random variable over many independent trials.

Variance and Standard Deviation

To measure how much a random variable deviates from its mean, we use its *variance*, denoted $\mathrm{Var}[X]$. This is defined as:

$$\mathrm{Var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

The *standard deviation* of X, denoted σ_X, is the square root of the variance:

$$\sigma_X = \sqrt{\mathrm{Var}[X]}.$$

A smaller variance (or standard deviation) means the random variable's values tend to lie closer to its mean, whereas a larger variance indicates a broader spread of values.

Asymptotic Notation

In describing the time complexity of algorithms, we use $\mathcal{O}(\ldots)$ notation in a standard way:

$$f(n) = \mathcal{O}(g(n)) \quad \Longleftrightarrow \quad \lim_{n\to\infty} \frac{f(n)}{g(n)} < \infty.$$

We may also use $\Theta(\ldots)$, $\Omega(\ldots)$, and other asymptotic notations where necessary.

Chebyshev's Inequality

Chebyshev's inequality is a cornerstone result in probability theory, providing a quantitative bound on how a random variable deviates from its expected value. Formally, let X be any random variable with finite mean $\mathbb{E}[X]$ and finite variance $\mathrm{Var}[X]$. Then, for any $k > 0$,

$$\mathbb{P}[|X - \mathbb{E}[X]| \geq k] \;\leq\; \frac{\mathrm{Var}[X]}{k^2}.$$

Intuition and Relation to Markov's Inequality

To build intuition, recall the simpler *Markov's inequality*, which states that for any nonnegative random variable Y and $t > 0$,

$$\mathbb{P}[Y \geq t] \leq \frac{\mathbb{E}[Y]}{t}.$$

Chebyshev's inequality is often viewed as an application of Markov's inequality to the random variable $(X - \mathbb{E}[X])^2$. Indeed,

$$\mathbb{P}[(X - \mathbb{E}[X])^2 \geq k^2] \leq \frac{\mathbb{E}[\left(X - \mathbb{E}[X]\right)^2]}{k^2} = \frac{\mathrm{Var}[X]}{k^2},$$

and from $\{\, |X - \mathbb{E}[X]| \geq k \,\} \;\equiv\; \{\, (X - \mathbb{E}[X])^2 \geq k^2 \,\}$, we obtain Chebyshev's inequality.

The key takeaway is that Chebyshev's inequality captures a broad class of distributions. It requires no assumption of normality or other specific distributional shapes, only finite variance. In exchange for this generality, Chebyshev's bound can be somewhat loose compared to more specialized inequalities (e.g., Hoeffding's, Chernoff's, or Bernstein's). Nonetheless, it remains a powerful and classic tool for bounding "tail events," i.e. events in which X significantly deviates from its mean.

Why We Use Chebyshev's Inequality Here

In our randomized median-finding algorithm, we draw random samples from the input array and use them to approximate the true median. The "rank" of our chosen pivots in the sorted order of the entire array are themselves random variables. We want to show that, with high probability, these rank containt the $\lceil n/2 \rceil$-th position.

By applying Chebyshev's inequality, we obtain a bound on $\mathbb{P}[|X - \mathbb{E}[X]| \geq k]$ for a well-chosen k, where X is the rank of our pivot. In other words, Chebyshev's inequality formally establishes the reliability of our sampling strategy by demonstrating that, except with small probability, our pivots partition the array around the actual median. This allows us to restrict further computation to a subset of the array whose size is guaranteed to be sufficiently small, thereby ensuring linear running time.

Extensions and Alternatives

While Chebyshev's inequality is sufficient for our purposes, more refined analyses might employ other concentration bounds such as Hoeffding's or Chernoff's inequalities. These stronger bounds can yield tighter probabilities when the underlying sample distribution has additional regularity properties (e.g., bounded random variables or independent Bernoulli trials). However, Chebyshev's inequality has the advantage of simplicity and broad applicability, making it an ideal choice for illustrating the core ideas in a clear manner.

Chapter 4

Algorithm and Analysis

In this chapter, we present our main randomized median-finding algorithm in full detail and analyze its performance. We first give a high-level description and provide the pseudocode in Algorithm 1. We then prove two central theorems: one that establishes the algorithm's probability of success (i.e. returning the correct median) and another that bounds its running time by $\mathcal{O}(n)$. Throughout this chapter, we use the notation introduced in Chapter 3.

Algorithm Description

The guiding idea behind the algorithm is to select a "small" random sample from the input array, sort this sample, and use carefully chosen elements from this sample as *pivots* to bracket the true median. Specifically:

- We draw $m = \lceil n^{3/4} \rceil$ elements uniformly at random from the input array $a[1 \ldots n]$ and sort these sampled elements into an array $b[1 \ldots m]$.
- We then pick two elements from b, denoted p_ℓ and p_r, near the middle of b but offset by $\pm\sqrt{n}$. Intuitively, these two pivots should "sandwich" the true median of the entire array with high probability.
- We gather all elements of the original array a that lie between these two pivots (inclusive) into a temporary array T. We verify that the median must lie in this sub-array, and that T is not excessively large.
- If those checks are satisfied, we sort T and extract the median; otherwise, the algorithm reports a *failure*.

The rest of this chapter gives a proof of two properties of this algorithm:

1. With high probability, Algorithm 1 does not fail and returns the correct median.
2. Conditioned on success, the overall running time is $\mathcal{O}(n)$.

Probability of Success

We begin by showing that, except with small probability, the two pivots p_ℓ and p_r are chosen so that the sub-array T contains the true median of a and is not excessively large. Formally, we have:

Theorem 4.1 (Correctness with High Probability) *The probability that Algorithm 1 returns the correct median is at least* $1 - \mathcal{O}\left(n^{-1/4}\right)$. *In other words, the algorithm succeeds* with high probability.

Algorithm 1 RandMedian: Randomized Median Algorithm

Require: Unsorted array $a[1 \ldots n]$ of size n.
Ensure: Median of a or *failure*.

Initialize an empty array b.
For each element $x \in a[1 \ldots n]$:
  Include x in b **with probability** $n^{-1/4}$, independently of other elements.
$m \leftarrow |b|$ ▷ (Random) size of the sample; note that $\mathbb{E}[m] = n^{3/4}$.
if $m > n^{3/4} + \sqrt{n}$ **then**
  return *failure*
end if
Sort b in $\mathcal{O}(m \log m)$ time
Compute pivots:

$$p_\ell \leftarrow b\left[\left\lfloor \frac{m}{2} - \sqrt{n} \right\rfloor\right], \quad p_r \leftarrow b\left[\left\lceil \frac{m}{2} + \sqrt{n} \right\rceil\right]$$

Define the subset $T \leftarrow \{\, x \in a : p_\ell \leq x \leq p_r \}$
Compute:

$$\begin{aligned} s &\leftarrow \#\{\, x \in a : x < p_\ell \}, \\ t &\leftarrow \#\{\, x \in a : p_\ell \leq x \leq p_r \}. \end{aligned}$$

if $s < \lceil n/2 \rceil$ **and** $s + t \geq \lceil n/2 \rceil$ **and** $t < 4n^{3/4}$ **then**
  Sort T in $\mathcal{O}(|T| \log |T|)$ time
  Median of $a \leftarrow T[\lceil n/2 \rceil - s]$
  return Median of a
else
  return *failure*
end if

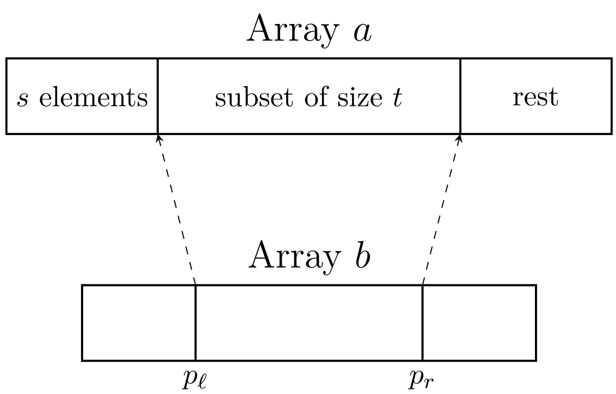

Figure 4.1: Illustration of the arrays and the pivots, showing array a, sampled array b, and the pivots p_ℓ, p_r. Note that this visualization shows the sorted version of the arrays.

Proof For simplicity, we assume throughout this proof that n is even and that $n^{3/4}$ is an integer; this avoids cumbersome floor and ceiling functions. The general case follows by analogous, but slightly more tedious, arguments.

Recall that $a[1 \dots n]$ is our input array of n distinct elements. Let $b[1 \dots m]$ be the sample of size m drawn by sampling each element with probability $n^{-1/4}$ from a. The setup is also visualized in Figure 4.1. We then sort b, pick the pivots[1]

$$p_\ell \;=\; b\Big[\tfrac{m}{2} - \sqrt{n}\Big], \quad p_r \;=\; b\Big[\tfrac{m}{2} + \sqrt{n}\Big],$$

and collect into T all elements $x \in a$ with $p_\ell \le x \le p_r$. Let

$$s \;=\; \#\{\, x \in a : x < p_\ell \,\} \quad \text{and} \quad t \;=\; \#\{\, x \in a : p_\ell \le x \le p_r \,\}.$$

Key sets and random variables. For any integer $k \in \{0, \dots, n\}$, define

$$S_k \;=\; \{\text{the } k \text{ smallest elements of } a\}.$$

We then define the random variable

$$X_k \;=\; \#\{\, x \in b : x \in S_k \}.$$

Since each element of a is independently included in b with probability $n^{-1/4}$, the number of elements of S_k that appear in b follows

$$X_k \;\sim\; \mathrm{Bin}\big(k,\; n^{-1/4}\big).$$

Hence,

$$\mathbb{E}[X_k] \;=\; k\, n^{-1/4}, \quad \mathrm{Var}[X_k] \;=\; k\, n^{-1/4}\big(1 - n^{-1/4}\big) \;\le\; k\, n^{-1/4}.$$

Application of Chebyshev's Inequality. Since $X_k \sim \mathrm{Bin}(k, n^{-1/4})$ with $\mathrm{Var}[X_k] \le k\, n^{-1/4}$, Chebyshev's inequality gives

$$\mathbb{P}[|X_k - k\, n^{-1/4}| \;\ge\; \sqrt{n}] \;\le\; \frac{\mathrm{Var}[X_k]}{n} \;\le\; \frac{k\, n^{-1/4}}{n} \;=\; \frac{k}{n^{5/4}}. \tag{4.1}$$

Defining the "bad" events. We show that the only way the algorithm can fail (i.e. not return the correct median) is if one of the following four events occurs. Denote

$$\begin{aligned}
A &:= \{\, s \;\ge\; n/2 \}\,, \\
B &:= \{\, s + t < n/2 \}\,, \\
C &:= \{\, s < \tfrac{n}{2} - 2\, n^{3/4} \}\,, \\
D &:= \{\, s + t \;\ge\; \tfrac{n}{2} + 2\, n^{3/4} \}\,. \\
E &:= \{\, m > n^{3/4} + \sqrt{n} \}\,.
\end{aligned}$$

Notice that E not occuring means that the first "return failure" line is not executed. If none of these events happens, then we must have

$$s < \tfrac{n}{2}, \quad s + t \;\ge\; \tfrac{n}{2}, \quad s \;\ge\; \tfrac{n}{2} - 2\, n^{3/4}, \quad \text{and} \quad s + t < \tfrac{n}{2} + 2\, n^{3/4}.$$

One can check that these conditions imply exactly

$$\big(s < \tfrac{n}{2}\big) \quad \text{and} \quad \big(s + t \ge \tfrac{n}{2}\big) \quad \text{and} \quad t < 4\, n^{3/4},$$

so the algorithm's final check (line 11 of the pseudocode) succeeds. In that case, our pivot-based bracket $[\, p_\ell,\, p_r \,]$ indeed contains the median, and the algorithm returns the correct value.

[1] Technically, if m is too small, the following lines are undefined (out of bounds). But this is avoided whp, as for the other events later in the proof.

Probability of each bad event. We next prove that $\mathbb{P}[A], \mathbb{P}[B], \mathbb{P}[C], \mathbb{P}[D], \mathbb{P}[E]$ are each at most $n^{-1/4}$. Observe:

- Event A means $s \geq n/2$. Hence among the $n/2$ smallest elements of a (i.e. the set $S_{n/2}$), at most $\frac{1}{2}\,n^{3/4} - \sqrt{n}$ of them are drawn into b. In other words,

$$X_{n/2} \;\leq\; \tfrac{1}{2}\,n^{3/4} \;-\; \sqrt{n}.$$

  But from $\mathbb{E}[X_{n/2}] = (n/2)\,n^{-1/4} = \frac{1}{2}\,n^{3/4}$ and (4.1), we get $\mathbb{P}[X_{n/2} \leq \frac{1}{2}\,n^{3/4} - \sqrt{n}] \leq n^{-1/4}$. Thus $\mathbb{P}[A] \leq n^{-1/4}$.

- Event B means $s + t < n/2$. Equivalently, the set $S_{n/2}$ has more than $\frac{1}{2}\,n^{3/4} + \sqrt{n}$ elements contained in b. Hence

$$X_{n/2} \;\geq\; \tfrac{1}{2}\,n^{3/4} \;+\; \sqrt{n}.$$

  Again, from $\mathbb{E}[X_{n/2}] = \frac{1}{2}\,n^{3/4}$ and (4.1), we get $\mathbb{P}[X_{n/2} \geq \frac{1}{2}\,n^{3/4} + \sqrt{n}] \leq n^{-1/4}$. So $\mathbb{P}[B] \leq n^{-1/4}$.

- Event C means $s < n/2 - 2\,n^{3/4}$. In this case, the set $S_{n/2-2\,n^{3/4}}$ must have at least $\frac{1}{2}\,n^{3/4} - \sqrt{n}$ of its elements in b. Formally,

$$X_{n/2-2\,n^{3/4}} \;\geq\; \tfrac{1}{2}\,n^{3/4} - \sqrt{n}.$$

  Using $\mathbb{E}[X_{n/2-2\,n^{3/4}}] = (n/2 - 2\,n^{3/4})\,n^{-1/4} = \frac{1}{2}n^{3/4} - 2\sqrt{n}$ and (4.1) again shows $\mathbb{P}[C] \leq n^{-1/4}$.

- Event D means $s + t \geq n/2 + 2\,n^{3/4}$. Hence the set $S_{n/2+2\,n^{3/4}}$ contains at most $\frac{1}{2}\,n^{3/4} + \sqrt{n}$ elements from b. That is,

$$X_{n/2+2\,n^{3/4}} \;\leq\; \tfrac{1}{2}\,n^{3/4} \;+\; \sqrt{n}.$$

  A final application of (4.1) yields $\mathbb{P}[D] \leq n^{-1/4}$.

- A similar calculation involving X_n shows that $\mathbb{P}[E] \leq n^{-1/4}$.

Conclusion. By the union bound,

$$\mathbb{P}[A \,\cup\, B \,\cup\, C \,\cup\, D \cup\, E] \;\leq\; \mathbb{P}[A] + \mathbb{P}[B] + \mathbb{P}[C] + \mathbb{P}[D] + \mathbb{P}[E] \;\leq\; 5\; \times n^{-1/4}$$

In other words, with probability at least $1 - 5n^{-1/4}$, *none* of these events occurs. When all five do not occur, the algorithm's checks

$$s < \lceil n/2 \rceil, \quad s + t \geq \lceil n/2 \rceil, \quad t < 4\,n^{3/4}$$

are satisfied, so the algorithm terminates successfully with the correct median. Hence the overall success probability is at least $1 - 5n^{-1/4}$, completing the proof. □

Runtime Analysis

Next, we establish that Algorithm 1 achieves linear time complexity *with high probability.* Note that if the algorithm returns *failure*, it may be retried or combined with a fallback procedure. Here, we focus on the typical execution that succeeds.

Theorem 4.2 (Runtime) *Conditioned on the event that Algorithm 1 does not fail (an event which, by Theorem 4.1, holds with high probability), the total running time is* $\mathcal{O}(n)$*.*

Proof We break the cost into four parts:

1. **Sampling and Sorting** b**.** We pick $m \leq n^{3/4} + \sqrt{n} \leq 2n^{3/4}$ elements randomly from a (conditioned on success). This is done in $\mathcal{O}(n)$ time. Sorting b of size m then takes

$$\mathcal{O}(m \log(m)) = \mathcal{O}\Big(n^{3/4} \log\Big(n^{3/4}\Big)\Big).$$

   Since $n^{3/4} \log\Big(n^{3/4}\Big)$ is asymptotically smaller than n for large n, this component is $o(n)$.

2. **Computing Ranks and Sub-array** T**.** We determine how many elements of a lie below p_ℓ, how many lie above p_r, etc. This amounts to a single pass through a, comparing each element of a to p_ℓ and p_r. Hence, it costs $\mathcal{O}(n)$ time.

3. **Sorting** T**.** By Algorithm 1, we only proceed to sort T if $t < 4n^{3/4}$. Thus,

$$t \leq 4n^{3/4}.$$

   Sorting T via a standard comparison-based algorithm costs

$$\mathcal{O}(|T| \log(|T|)) \leq \mathcal{O}\Big(n^{3/4} \log\Big(\big(n^{3/4}\big)\Big)\Big),$$

   which again is $o(n)$.

4. **Extracting the Median.** After sorting T, we simply pick the element of T whose rank corresponds to $\lceil n/2 \rceil - s$. This is clearly an $\mathcal{O}(1)$ operation.

Summing these components, the *dominant term* is $\mathcal{O}(n)$ (from the pass to identify s and t). The sublinear sorting costs on b and on T both constitute $\mathcal{O}\Big(n^{3/4} \log(n)\Big)$, which is strictly smaller than n for large n. Consequently, the total running time is $\mathcal{O}(n)$. □

In conclusion, the previous two theorems immediately prove Theorem 1.1, our main result, which we restate here for convenience.

Theorem 1.1 *There exists a randomized algorithm to compute the median that runs in time* $\mathcal{O}(n)$ *and succeeds with probability* $1 - \mathcal{O}\Big(n^{-1/4}\Big)$*.*

Discussion and Extensions

Theorems 4.1 and 4.2 jointly show that Algorithm 1 achieves $\mathcal{O}(n)$ time with high probability of success. In rare cases where the check fails, the algorithm returns *failure* rather than risking an incorrect answer. One can either re-run the algorithm (the probability of repeated failures being very low) or combine it with a fallback deterministic linear-time median-finding routine such as the median-of-medians approach. The randomized algorithm typically attains lower hidden constants in practice than purely deterministic linear-time solutions, thereby offering an attractive balance of theoretical guarantees and real-world performance.

In the next chapter, we summarize our results and highlight possible directions for future work on randomized selection algorithms.

Chapter 5

Model-free multi-currency portfolio returns

In this chapter, we derive a model-free representation of one-period portfolio returns for an investor who holds assets denominated in multiple currencies and can take currency forward positions for hedging. The derivations follow the methodology presented in [**?**] and [**?**], using simple returns and assume that rebalancing occurs at the end of each period. The representation is used in subsequent chapters to formulate and analyze the results of the multi-currency portfolio optimization problem.

5.1 Unhedged Portfolio Returns

We consider an international investor that allocates his wealth across N assets denominated in up to $M+1$ currencies. We fix a discrete time index $t \in \{0, 1, 2, \dots\}$ and one investment period from t to $t{+}1$. Let the investor's *domestic currency* be indexed by 0 and let the rest of the M foreign currencies be indexed by $\mathcal{C} \coloneqq \{1, 2, \dots, M\}$.

Let $P_{i,t} > 0$ be the price of asset i at time t in its local currency $c_i \in \mathcal{C} \cup \{0\}$. Define the corresponding *simple local currency asset return* over $(t, t{+}1]$ as

$$R^{\mathrm{LC}}_{i,t+1} \coloneqq \frac{P_{i,t+1} - P_{i,t}}{P_{i,t}}.$$

Thus $P_{i,t+1} = P_{i,t}(1 + R^{\mathrm{LC}}_{i,t+1})$.

For each foreign currency $c \in \mathcal{C}$, let $S_{c,t} > 0$ denote the exchange rate at time t, quoted as units of domestic currency per one unit of foreign currency c. The associated *simple currency return* over $(t, t{+}1]$ is

$$e_{c,t+1} \coloneqq \frac{S_{c,t+1} - S_{c,t}}{S_{c,t}}.$$

Equivalently, $S_{c,t+1} = S_{c,t}(1 + e_{c,t+1})$. By definition for the domestic currency, we set $S_{0,t} \equiv 1$ and $e_{0,t+1} \equiv 0$.

The domestic currency value of holding one unit of asset i at time t is $P_{i,t}\, S_{c_i,t}$. Therefore, the *unhedged domestic currency simple return* of asset i is

$$\begin{aligned}
R^{u}_{i,t+1} &= \frac{P_{i,t+1} S_{c_i,t+1} - P_{i,t} S_{c_i,t}}{P_{i,t} S_{c_i,t}} \\
&= \frac{P_{i,t}(1 + R^{\mathrm{LC}}_{i,t+1}) S_{c_i,t}(1 + e_{c_i,t+1}) - P_{i,t} S_{c_i,t}}{P_{i,t} S_{c_i,t}} \\
&= R^{\mathrm{LC}}_{i,t+1} + e_{c_i,t+1} + R^{\mathrm{LC}}_{i,t+1}\, e_{c_i,t+1}. \qquad (5.1)
\end{aligned}$$

Let $\boldsymbol{x}_t \in \mathbb{R}^N$ be the vector of asset portfolio weights chosen at time t and held until $t{+}1$, where $x_{i,t}$ is the fraction of total portfolio value invested in asset i at time t. We impose the standard budget constraint $\mathbf{1}^\mathsf{T}\boldsymbol{x}_t = 1$.

Assuming no rebalancing between times t and $t{+}1$, the domestic currency portfolio's unhedged return is given by

$$R^u_{\mathcal{P},t+1} = \sum_{i=1}^{N} x_{i,t}\, R^u_{i,t+1}.$$

5.2 Hedged Portfolio Returns

To manage currency risk, the investor can take positions in one-period currency forward contracts. A forward contract on foreign currency c entered at time t settles at time $t{+}1$ with payoff (in domestic currency) of $F_{c,t} - S_{c,t+1}$, where $F_{c,t}$ is the forward rate quoted at time t. The forward rate is typically set such that the contract has zero value at inception, i.e. $F_{c,t} = S_{c,t}(1+r_t)/(1+r_{c,t})$, where r_t and $r_{c,t}$ are the domestic and foreign risk-free rates, respectively. The *excess currency return* over the period is $e_{c,t+1} - f_{c,t}$, where $f_{c,t} = (F_{c,t} - S_{c,t})/S_{c,t}$ is the forward premium. The investor can choose the forward position $\tau_{c,t}$ (expressed as a fraction of total portfolio value) for each foreign currency c to hedge against currency risk. The *hedged* portfolio return is then the sum of the unhedged asset returns and the payoffs from the forward positions:

$$R^h_{\mathcal{P},t+1} = \sum_{i=1}^{N} x_{i,t}\, R^u_{i,t+1} + \sum_{c=1}^{M} \tau_{c,t}\,(f_{c,t} - e_{c,t+1}).$$

For each foreign currency $c \in \mathcal{C}$, assume that at time t the investor can enter a one-period currency forward contract that settles at $t{+}1$, with forward rate $F_{c,t}$ quoted in domestic currency per unit of foreign currency c. Define the *forward premium* (simple rate) by

$$f_{c,t} \coloneqq \frac{F_{c,t} - S_{c,t}}{S_{c,t}}.$$

Let $\boldsymbol{\tau}_t \in \mathbb{R}^M$ denote forward positions expressed as fractions of total portfolio value, where $\tau_{c,t}$ represents the relative notional of the *short* forward in currency c at time t. With this sign convention, $\tau_{c,t} > 0$ represents hedging (shorting foreign currency forward). The *hedged* portfolio return is

$$R^h_{P,t+1} = \sum_{i=1}^{N} x_{i,t}\, R^u_{i,t+1} + \sum_{c=1}^{M} \varphi_{c,t}\,(f_{c,t} - e_{c,t+1}). \tag{5.2}$$

This representation is model-free: it holds pathwise for realized returns and does not assume any distributional structure.

5.3 Matrix Representation and Decomposition

To express the returns in a compact form suitable for optimization, we define the currency incidence matrix $\boldsymbol{C} \in \{0,1\}^{N\times M}$ by

$$C_{i,c} \coloneqq \begin{cases} 1, & \text{if asset } i \text{ is denominated in foreign currency } c, \\ 0, & \text{otherwise.} \end{cases}$$

The portfolio's *implicit foreign-currency exposure* is $\boldsymbol{w}_t \coloneqq \boldsymbol{C}^\mathsf{T}\boldsymbol{x}_t \in \mathbb{R}^M$, where $w_{c,t} = \sum_{i:\, c_i = c} x_{i,t}$.

Collect returns into vectors $\boldsymbol{R}_{t+1}^{\mathrm{LC}} \in \mathbb{R}^N$, $\boldsymbol{e}_{t+1} \in \mathbb{R}^M$, and the forward premiums $\boldsymbol{f}_t \in \mathbb{R}^M$. Define the currency-return vector aligned to assets via $\boldsymbol{C}\boldsymbol{e}_{t+1} \in \mathbb{R}^N$, where the ith component is $e_{c_i,t+1}$ for foreign assets and 0 for domestic assets. Using (5.1), the vector of *unhedged domestic returns* across assets is

$$\boldsymbol{R}_{t+1}^{u} = \boldsymbol{R}_{t+1}^{\mathrm{LC}} + \boldsymbol{C}\boldsymbol{e}_{t+1} + (\boldsymbol{R}_{t+1}^{\mathrm{LC}} \odot \boldsymbol{C}\boldsymbol{e}_{t+1}),$$

where $\odot$ denotes the Hadamard product.

Chapter 6

Empirical Analysis

In this chapter, we present an empirical analysis of the multi-currency portfolio optimization problem using historical data. We apply the model-free return representation derived in the previous chapter to compute realized returns for various portfolio strategies. We evalutate the performance of the various strategies on two investment universes, a large investment universe consisting of 50 biggest stocks in the 14 largest developed markets and a small investment universe consisting of 10 biggest stocks in the same markets. We compare the performance of the optimized portfolios to a benchmark equally weighted portfolio.

6.1 Data

Our analysis relies on a comprehensive dataset of international equities, foreign exchange spot rates, and forward exchange rates sourced from Refinitiv Datastream. The dataset covers a period from June 1990 to June 2023, capturing various market regimes, including the dot-com bubble burst, the 2008 Global Financial Crisis, the European debt crisis, and the COVID-19 pandemic.

We construct our investment universe from the 14 largest developed equity markets: Australia, Canada, France, Germany, Hong Kong, Italy, Japan, the Netherlands, Singapore, Spain, Sweden, Switzerland, the United Kingdom, and the United States. To construct the local equity portfolios, we select the largest stocks in each market based on their market capitalization at the beginning of each year. We consider two distinct cross-sections of stocks to analyze the impact of the investment universe size on portfolio performance, a small universe consisting of the top 10 stocks per market (140 stocks in total) and a large universe consisting of the top 50 stocks per market (700 stocks in total).

For each selected stock, we retrieve daily total return indices, which account for dividend reinvestments, and compute the corresponding local currency returns. The US dollar serves as the base currency for our analysis, we calculate the fully hedged returns for each stock following the model-free return representation in chapter 5.

6.2 Results: Small Investment Universe

6.3 Results: Large Investment Universe

Chapter 7

Conclusion

Appendices

Appendix A

ES Optimization Equivalence

We derive the equivalence between the direct minimization of Expected Shortfall and the auxiliary formulation proposed by Rockafellar and Uryasev (2000) that allows for formulating the optimization as a linear program.

We assume that the loss distribution is continuous, meaning the loss cumulative distribution function has no jumps and the probability density function $f(l)$ exists. Under these assumptions, Value-at-Risk (VaR) is uniquely defined.

Define the loss random variable at time $t+1$ given portfolio weights x_t as $L(x_t) = -x_t^T R_{t+1}$. Let $f(l)$ denote the probability density function of the loss $L(x_t)$. The VaR at confidence level β is defined as the β-quantile of the loss distribution

$$\mathrm{VaR}_\beta(x_t) = \min\{\alpha : \mathbb{P}(L(x_t) \leq \alpha) \geq \beta\}.$$

The Expected Shortfall is defined as the conditional expectation of losses exceeding VaR

$$\mathrm{ES}_\beta(x_t) = \mathbb{E}[L(x_t) \mid L(x_t) \geq \mathrm{VaR}_\beta(x_t)] = \frac{1}{1-\beta} \int_{\mathrm{VaR}_\beta(x_t)}^{\infty} l f(l)\, dl.$$

We now consider the auxiliary function F_β defined as

$$F_\beta(x_t, \alpha) = \alpha + \frac{1}{1-\beta} \mathbb{E}_t[(L(x_t) - \alpha)^+].$$

We can rewrite the expectation term using the integral form

$$\mathbb{E}_t[(L(x_t) - \alpha)^+] = \int_{-\infty}^{\infty} (l - \alpha)^+ f(l)\, dl = \int_{\alpha}^{\infty} (l - \alpha) f(l)\, dl.$$

Before minimizing $F_\beta(x_t, \alpha)$ with respect to α, we verify convexity. The second derivative of F_β with respect to α is

$$\begin{aligned} \frac{\partial F_\beta}{\partial \alpha} &= 1 - \frac{1}{1-\beta} \int_{\alpha}^{\infty} f(l)\, dl \\ \frac{\partial^2 F_\beta}{\partial \alpha^2} &= \frac{1}{1-\beta} f(\alpha). \end{aligned}$$

Since the probability density function $f(\alpha) \geq 0$ and $\beta < 1$, the second derivative is non-negative, $\frac{\partial^2 F_\beta}{\partial \alpha^2} \geq 0$. Thus, $F_\beta(x_t, \alpha)$ is convex in α, ensuring that the first-order condition is necessary and sufficient for a global minimum.

Differentiation of $F_\beta(x_t,\alpha)$ with respect to α yields the first-order condition for optimality. Using Leibniz's rule we have

$$\begin{aligned}
\frac{\partial F_\beta}{\partial \alpha} &= 1 + \frac{1}{1-\beta}\frac{\partial}{\partial \alpha}\int_\alpha^\infty (l-\alpha) f(l)\, dl \\
&= 1 + \frac{1}{1-\beta}\left[-f(\alpha)\cdot 0 + \int_\alpha^\infty (-1) f(l)\, dl\right] \\
&= 1 - \frac{1}{1-\beta}\int_\alpha^\infty f(l)\, dl \\
&= 1 - \frac{1}{1-\beta}\mathbb{P}(L(x_t) \geq \alpha).
\end{aligned}$$

Setting the derivative to zero to find the optimal α^*

$$\begin{aligned}
1 - \frac{1}{1-\beta}\mathbb{P}(L(x_t) \geq \alpha^*) &= 0 \\
\mathbb{P}(L(x_t) \geq \alpha^*) &= 1-\beta \\
\mathbb{P}(L(x_t) \leq \alpha^*) &= \beta.
\end{aligned}$$

This implies that the optimal α^* that minimizes the auxiliary function for a fixed x_t is exactly the Value-at-Risk at level β, so $\alpha^* = \mathrm{VaR}_\beta(x_t)$.

Substituting $\alpha^* = \mathrm{VaR}_\beta(x_t)$ back into $F_\beta(x_t,\alpha)$ we obtain

$$\begin{aligned}
F_\beta(x_t, \mathrm{VaR}_\beta(x_t)) &= \mathrm{VaR}_\beta(x_t) + \frac{1}{1-\beta}\int_{\mathrm{VaR}_\beta(x_t)}^\infty (l - \mathrm{VaR}_\beta(x_t)) f(l)\, dl \\
&= \mathrm{VaR}_\beta(x_t) + \frac{1}{1-\beta}\left(\int_{\mathrm{VaR}_\beta(x_t)}^\infty l f(l)\, dl - \mathrm{VaR}_\beta(x_t)\int_{\mathrm{VaR}_\beta(x_t)}^\infty f(l)\, dl\right).
\end{aligned}$$

Recognizing that $\int_{\mathrm{VaR}_\beta(x_t)}^\infty f(l)\, dl = \mathbb{P}(L(x_t) \geq \mathrm{VaR}_\beta(x_t)) = 1-\beta$, we have

$$\begin{aligned}
F_\beta(x_t, \mathrm{VaR}_\beta(x_t)) &= \mathrm{VaR}_\beta(x_t) + \frac{1}{1-\beta}\int_{\mathrm{VaR}_\beta(x_t)}^\infty l f(l)\, dl - \frac{1}{1-\beta}\mathrm{VaR}_\beta(x_t)(1-\beta) \\
&= \mathrm{VaR}_\beta(x_t) + \mathrm{ES}_\beta(x_t) - \mathrm{VaR}_\beta(x_t) \\
&= \mathrm{ES}_\beta(x_t).
\end{aligned}$$

Thus, minimizing $\mathrm{ES}_\beta(x_t)$ with respect to x_t is equivalent to minimizing $F_\beta(x_t,\alpha)$ jointly with respect to x_t and α

$$\min_{x_t} \mathrm{ES}_\beta(x_t) = \min_{x_t,\alpha}\left(\alpha + \frac{1}{1-\beta}\mathbb{E}_t[(-x_t^T R_{t+1} - \alpha)^+]\right).$$

Finally, we observe that $F_\beta(x_t,\alpha)$ is jointly convex in (x_t,α). The term α is linear (hence convex). The inner map $(x_t,\alpha) \mapsto (-x_t^\top R_{t+1} - \alpha)$ is affine. The outer function $z \mapsto z^+$ is convex and nondecreasing. The composition of a convex nondecreasing function with an affine map is convex. Thus $(-x_t^\top R_{t+1} - \alpha)^+$ is jointly convex in (x_t,α). Taking the expectation preserves convexity.

Since the objective function is convex, any local minimum is global, ensuring that optimization algorithms will converge efficiently to the optimal solution.

Appendix B

ES Optimization Implementation

Building upon the continuous equivalence derived in Appendix A and following the optimization framework in the original paper (Ulyrch, Lucescu 2026), this section details the precise matrix formulations required to solve the regularized multi-currency Expected Shortfall models. By linearizing the L_1 penalty and discretizing the ES computations, we map both the separate overlay and joint optimization frameworks into standard Quadratic Programs of the form

$$\min_{\tilde{z}} \left(\tilde{c}^T \tilde{z} + \frac{1}{2} \tilde{z}^T \tilde{Q} \tilde{z} \right) \quad \text{subject to} \quad \tilde{D}\tilde{z} = \tilde{E}, \quad \tilde{A}\tilde{z} \leq b, \quad \tilde{z} \geq 0, \tag{B.1}$$

where $\tilde{z}$ is an augmented decision vector containing the portfolio weights and all necessary auxiliary variables.

B.1 Sample-Based Expected Shortfall

In practice, the conditional expectation from the ES objective is approximated using q historical joint return observations. Let $R_{\text{data}} \in \mathbb{R}^{q \times N}$ be the matrix of historical asset returns, where each row R_k^T represents the return vector at time k. The sample-based objective becomes:

$$\min_{x_t, \alpha} \left(-\mu^T x_t + \gamma \alpha + \frac{\gamma}{q(1-\beta)} \sum_{k=1}^{q} (-R_k^T x_t - \alpha)^+ \right).$$

To linearize the maximum operator $(\cdot)^+$, Rockafellar and Uryasev (2000) introduce a q-dimensional vector of auxiliary variables $u \in \mathbb{R}^q$, where $u_k = (-R_k^T x_t - \alpha)^+$. This implies two conditions for each sample k:

$$\begin{aligned} u_k &\geq 0 \\ u_k &\geq -R_k^T x_t - \alpha. \end{aligned}$$

Rearranging the second condition to isolate the variables yields a standard linear inequality:

$$-R_k^T x_t - \alpha - u_k \leq 0 \quad \forall\, k \in \{1, \ldots, q\}. \tag{B.2}$$

In block matrix notation across all q samples, this becomes:

$$-R_{\text{data}} x_t - \mathbf{1}_q \alpha - I_q u \leq \mathbf{0}_q,$$

where $\mathbf{1}_q$ is a q-dimensional vector of ones, I_q is the $q \times q$ identity matrix, and $\mathbf{0}_q$ is a q-dimensional vector of zeros. This constructs the first three block columns of the general inequality constraint matrix $\tilde{A}$.

B.2 Linearization of the L_1 Penalty and Leverage

The L_1 norm $\|x_t\|_1 = \sum_{i=1}^{N} |x_{i,t}|$ introduces non-differentiability into the objective function. To resolve this, we introduce non-negative auxiliary variables representing the positive and negative components of each asset weight: $\Delta_t^+, \Delta_t^- \in \mathbb{R}^N$ such that $\Delta_t^+ \geq 0$ and $\Delta_t^- \geq 0$.

The weights are decomposed as:

$$x_t = \Delta_t^+ - \Delta_t^- \implies I_N x_t + I_N \Delta_t^- - I_N \Delta_t^+ = \mathbf{0}_N. \tag{B.3}$$

Because the optimization minimizes the penalties, at most one of $\Delta_{i,t}^+$ or $\Delta_{i,t}^-$ will be non-zero for any asset i. Therefore, the absolute value is exactly determined by their sum: $|x_{i,t}| = \Delta_{i,t}^+ + \Delta_{i,t}^-$.

This allows the L_1 penalty term to be formulated linearly in the objective function as $\lambda_1 \mathbf{1}_N^T \Delta_t^+ + \lambda_1 \mathbf{1}_N^T \Delta_t^-$. Furthermore, the portfolio leverage constraint $\|x_t\|_1 \leq \ell$ transforms into a linear inequality:

$$\mathbf{1}_N^T \Delta_t^+ + \mathbf{1}_N^T \Delta_t^- \leq \ell. \tag{B.4}$$

B.3 Quadratic Translation of the L_2 Penalty

Unlike the expected shortfall and L_1 terms which map to the linear objective vector $\tilde{c}$, the L_2 penalty integrates naturally into the quadratic matrix $\tilde{Q}$. The penalty is defined as:

$$\lambda_2 \|x_t\|_2^2 = \lambda_2 x_t^T x_t = \frac{1}{2} x_t^T (2\lambda_2 I_N) x_t.$$

Thus, the block of $\tilde{Q}$ corresponding to the weights x_t is populated with the diagonal matrix $2\lambda_2 I_N$, while all auxiliary variables receive zeros.

B.4 Separate Optimization (Currency Overlay)

In the separate approach, asset weights $x_t \in \mathbb{R}^N$ and currency exposures $\psi_t \in \mathbb{R}^M$ are optimized sequentially.

B.4.1 Step 1: Asset-Level Optimization

The target vector relies on fully hedged asset returns $R^{\text{fh}} \in \mathbb{R}^{q \times N}$. We define the augmented vector of dimension $(3N + q + 1)$:

$$\tilde{x}_t = \left(x_t^T, \quad \alpha, \quad u^T, \quad (\Delta_t^-)^T, \quad (\Delta_t^+)^T \right)^T.$$

The objective vector $\tilde{c}$ and quadratic matrix $\tilde{Q}$ are constructed as:

$$\tilde{c} = \begin{pmatrix} -\mu_{\text{fh}} \\ \gamma \\ \frac{\gamma}{q(1-\beta)} \mathbf{1}_q \\ \lambda_1^a \mathbf{1}_N \\ \lambda_1^a \mathbf{1}_N \end{pmatrix}, \quad \tilde{Q} = \begin{pmatrix} 2\lambda_2^a I_N & O \\ O & O \end{pmatrix},$$

where O represents zero matrices of appropriate dimensions. The equality constraints enforce the budget condition $\mathbf{1}_N^T x_t = 1$ and the variable decomposition from Equation B.3:

$$\tilde{D} = \begin{pmatrix} \mathbf{1}_N^T & 0 & \mathbf{0}_q^T & \mathbf{0}_N^T & \mathbf{0}_N^T \\ I_N & \mathbf{0}_N & O_{N \times q} & I_N & -I_N \end{pmatrix}, \quad \tilde{E} = \begin{pmatrix} 1 \\ \mathbf{0}_N \end{pmatrix}.$$

The inequality matrix $\tilde{A}$ and bound vector b encode the shortfall losses from Equation B.2 and the leverage limit from Equation B.4:

$$\tilde{A} = \begin{pmatrix} -R^{\text{fh}} & -\mathbf{1}_q & -I_q & O_{q\times N} & O_{q\times N} \\ \mathbf{0}_N^T & 0 & \mathbf{0}_q^T & \mathbf{1}_N^T & \mathbf{1}_N^T \end{pmatrix}, \quad b = \begin{pmatrix} \mathbf{0}_q \\ \ell \end{pmatrix}.$$

B.4.2 Step 2: Currency-Level Optimization

Conditional on the optimal asset weights x_t^*, the currency exposures ψ_t are optimized. Using historical excess currency returns $R^c \in \mathbb{R}^{q\times M}$, the augmented vector becomes $\tilde{\psi}_t \in \mathbb{R}^{(3M+q+1)}$.

Because currency overlays act as zero-investment portfolios, there is no budget constraint. Additionally, the pre-fixed asset portfolio returns $(R^{\text{fh}} x_t^*)$ must be subtracted from the shortfall inequality constraint boundaries. Thus, the formulations shift to:

$$\tilde{D}_\psi = \begin{pmatrix} I_M & \mathbf{0}_M & O_{M\times q} & I_M & -I_M \end{pmatrix}, \quad \tilde{E}_\psi = \mathbf{0}_M,$$

$$\tilde{A}_\psi = \begin{pmatrix} -R^c & -\mathbf{1}_q & -I_q & O_{q\times M} & O_{q\times M} \end{pmatrix}, \quad b_\psi = R^{\text{fh}} x_t^*.$$

B.5 Joint Optimization Framework

The joint formulation merges assets and currencies into a single decision vector $\theta_t = (x_t^T, \psi_t^T)^T \in \mathbb{R}^{N+M}$, evaluated over the joint return block matrix $R^{\text{all}} = \begin{pmatrix} R^{\text{fh}} & R^c \end{pmatrix} \in \mathbb{R}^{q\times(N+M)}$.

We construct an expansive augmented vector $\tilde{\theta}_t \in \mathbb{R}^{(3N+3M+q+1)}$:

$$\tilde{\theta}_t = \begin{pmatrix} x_t^T, & \psi_t^T, & \alpha, & u^T, & (\Delta_{x,t}^-)^T, & (\Delta_{\psi,t}^-)^T, & (\Delta_{x,t}^+)^T, & (\Delta_{\psi,t}^+)^T \end{pmatrix}^T.$$

The objective vector applies separate L_1 penalty parameters (λ_1^a for assets, λ_1^c for currencies):

$$\tilde{c} = \begin{pmatrix} -\mu_{\text{fh}} \\ -\mu_c \\ \gamma \\ \frac{\gamma}{q(1-\beta)}\mathbf{1}_q \\ \lambda_1^a \mathbf{1}_N \\ \lambda_1^c \mathbf{1}_M \\ \lambda_1^a \mathbf{1}_N \\ \lambda_1^c \mathbf{1}_M \end{pmatrix}, \quad \tilde{Q} = \begin{pmatrix} 2\lambda_2^a I_N & O & O \\ O & 2\lambda_2^c I_M & O \\ O & O & O \end{pmatrix}.$$

The budget constraint applies strictly to the asset weights, while the decomposition constraints apply to both assets and currencies.

$$\tilde{D} = \begin{pmatrix} \mathbf{1}_N^T \quad \mathbf{0}_M^T & 0 & \mathbf{0}_q^T & \mathbf{0}_{N+M}^T & \mathbf{0}_{N+M}^T \\ I_{N+M} & \mathbf{0}_{N+M} & O_{(N+M)\times q} & I_{N+M} & -I_{N+M} \end{pmatrix}, \quad \tilde{E} = \begin{pmatrix} 1 \\ \mathbf{0}_{N+M} \end{pmatrix}.$$

Finally, the inequality matrix binds the unified shortfall distribution and enforces the leverage limit solely on the gross asset exposure:

$$\tilde{A} = \begin{pmatrix} -R^{\text{all}} & -\mathbf{1}_q & -I_q & O_{q\times(2N+2M)} & \\ \mathbf{0}_{N+M}^T & 0 & \mathbf{0}_q^T & \begin{pmatrix} \mathbf{1}_N^T & \mathbf{0}_M^T \end{pmatrix} & \begin{pmatrix} \mathbf{1}_N^T & \mathbf{0}_M^T \end{pmatrix} \end{pmatrix}, \quad b = \begin{pmatrix} \mathbf{0}_q \\ \ell \end{pmatrix}.$$

Solving this comprehensive quadratic program yields the optimal regularized asset and currency allocations concurrently.

Appendix C

Transaction Costs

Following the methodology outlined by [2], [2] and [?], and keeping the notation introduced in Chapter 5, we model transaction costs as proportional to the volume of traded assets and currencies, ignoring any fixed trading costs. The return of the hedged portfolio is expressed as the sum of unhedged asset returns and gains from forward currency positions

$$R^{h}_{P,t+1} = \sum_{i=1}^{N} x_{i,t} R^{u}_{i,t+1} + \sum_{c=1}^{M} \varphi_{c,t}(f_{c,t} - e_{c,t+1}). \tag{(5.2)}$$

Let τ^{asset}_{t+1} and τ^{curr}_{t+1} denote the total transaction costs associated with asset and currency trades, respectively. The portfolio return net of transaction costs is then given by

$$R^{h,net}_{P,t+1} = R^{h}_{P,t+1} - \tau^{asset}_{t+1} - \tau^{curr}_{t+1}. \tag{C.1}$$

The total transaction costs are computed as the sum of the costs incurred from trading individual assets and currencies, weighted by the respective trading volumes

$$\tau^{asset}_{t+1} = \sum_{i=1}^{N} \tau^{asset}_{i,t+1} |x_{i,t+1} - x^{-}_{i,t+1}|,$$

$$\tau^{curr}_{t+1} = \sum_{c=1}^{M} \tau^{curr}_{c,t+1} \varphi_{c,t+1} + \sum_{c=1}^{M} \tau^{curr}_{c,t+1} |w_{c,t+1} - w^{-}_{c,t+1}|,$$

where $\tau^{asset}_{i,t+1}$ and $\tau^{curr}_{c,t+1}$ are the per-unit transaction costs for asset i and currency c. The terms $x^{-}_{i,t+1}$ and $w^{-}_{c,t+1}$ denote the asset weights and implied currency weights immediately prior to rebalancing at time $t+1$ (i.e., the drifted weights).

Asset transaction costs are incurred exclusively when rebalancing individual asset holdings. Currency transaction costs, however, arise from two sources, adjustments in forward hedge positions (assuming the forward contract maturity matches the rebalancing frequency, the old contract expires and a new contract of full notional size $\varphi_{c,t+1}$ must be established at each rebalancing date) and currency exchanges required to fund purchases of assets denominated in different currencies ($|w_{c,t+1} - w^{-}_{c,t+1}|$). If the portfolio is rebalanced more frequently than the forward contract maturity, existing forward contracts are kept open and only incremental adjustments are traded. In this case, the forward transaction cost term is modeled as the absolute difference, replacing $\sum_{c=1}^{M} \tau^{curr}_{c,t+1} \varphi_{c,t+1}$ with $\sum_{c=1}^{M} \tau^{curr}_{c,t+1} |\varphi_{c,t+1} - \varphi^{-}_{c,t+1}|$, where $\varphi^{-}_{c,t+1}$ represents the drifted hedge weight. Following [2], these costs are aggregated at the currency level, operating under the assumption that proceeds from selling assets denominated in currency c can fund the purchase of other assets in the same currency without incurring currency conversion fees.

To define the individual per-unit costs, we consider an asset i denominated in currency c_i. Let $P_{i,t}^{ask}$, $P_{i,t}^{bid}$, and $P_{i,t}$ represent the ask, bid, and mid prices of the asset, while $S_{c_i,t}^{ask}$, $S_{c_i,t}^{bid}$, and $S_{c_i,t}$ denote the corresponding currency prices. If asset i is purchased at time t and sold at time $t+1$, its realized log return is given by

$$
\begin{aligned}
\log\left(\frac{P_{i,t+1}^{bid}S_{c_i,t+1}^{bid}}{P_{i,t}^{ask}S_{c_i,t}^{ask}}\right) &= \log\left(\frac{\left(P_{i,t+1}-\frac{\delta_{i,t+1}^{asset}}{2}\right)\left(S_{c_i,t+1}-\frac{\delta_{c_i,t+1}^{curr}}{2}\right)}{\left(P_{i,t}+\frac{\delta_{i,t}^{asset}}{2}\right)\left(S_{c_i,t}+\frac{\delta_{c_i,t}^{curr}}{2}\right)}\right) \\
&= \log\left(\frac{P_{i,t+1}\left(1-\frac{\delta_{i,t+1}^{asset}}{2P_{i,t+1}}\right)S_{c_i,t+1}\left(1-\frac{\delta_{c_i,t+1}^{curr}}{2S_{c_i,t+1}}\right)}{P_{i,t}\left(1+\frac{\delta_{i,t}^{asset}}{2P_{i,t}}\right)S_{c_i,t}\left(1+\frac{\delta_{c_i,t}^{curr}}{2S_{c_i,t}}\right)}\right) \\
&= \log\left(\frac{P_{i,t+1}S_{c_i,t+1}}{P_{i,t}S_{c_i,t}}\right)+\log\left(\frac{1-\frac{\delta_{i,t+1}^{asset}}{2P_{i,t+1}}}{1+\frac{\delta_{i,t}^{asset}}{2P_{i,t}}}\right)+\log\left(\frac{1-\frac{\delta_{c_i,t+1}^{curr}}{2S_{c_i,t+1}}}{1+\frac{\delta_{c_i,t}^{curr}}{2S_{c_i,t}}}\right) \\
&= \log\left(\frac{P_{i,t+1}S_{c_i,t+1}}{P_{i,t}S_{c_i,t}}\right)-\log\left(\frac{1+\frac{\delta_{i,t}^{asset}}{2P_{i,t}}}{1-\frac{\delta_{i,t+1}^{asset}}{2P_{i,t+1}}}\right)-\log\left(\frac{1+\frac{\delta_{c_i,t}^{curr}}{2S_{c_i,t}}}{1-\frac{\delta_{c_i,t+1}^{curr}}{2S_{c_i,t+1}}}\right),
\end{aligned}
$$

where δ_i^{asset} and $\delta_{c_i}^{curr}$ are the bid-ask spreads. Comparing this to equation (C.1), the transaction cost is precisely the difference between the actual realized log return and mid-price returs for the asset and currency.

Defining half of the proportional bid-ask spreads as

$$
\chi_{i,t}^{asset} = \frac{P_{i,t}^{ask}-P_{i,t}^{bid}}{2P_{i,t}} \quad \text{and} \quad \chi_{c,t}^{curr} = \frac{S_{c,t}^{ask}-S_{c,t}^{bid}}{2S_{c,t}},
$$

the the per-unit transaction costs for trading asset i and currency c become

$$
\tau_{i,t+1}^{asset} = \begin{cases} \log\left(\frac{1+\chi_{i,t+1}^{asset}}{1-\chi_{i,t}^{asset}}\right) & \text{if } x_{i,t+1} > x_{i,t+1}^{-}, \\ \log\left(\frac{1-\chi_{i,t}^{asset}}{1+\chi_{i,t+1}^{asset}}\right) & \text{otherwise,} \end{cases}
$$

and

$$
\tau_{c,t+1}^{curr} = \begin{cases} \log\left(\frac{1+\chi_{c,t+1}^{curr}}{1-\chi_{c,t}^{curr}}\right) & \text{if } \varphi_{c,t+1} > 0 \text{ or } w_{c,t+1} > w_{c,t+1}^{-}, \\ \log\left(\frac{1-\chi_{c,t}^{curr}}{1+\chi_{c,t+1}^{curr}}\right) & \text{otherwise.} \end{cases}
$$

For our empirical analysis, we adopt the simplifying assumption, that proportional bid-ask spreads remain constant both over time and across the cross-sections of both assets and currencies. We set the half-spreads to $\chi^{asset} = 0.001$ (10 bps) for all equities and $\chi^{curr} = 0.0001$ (1 bp) for all currencies, consistent with empirical finidngs in [**?**] and [**?**]. Under these assumptions, the transaction costs simplify to constant values of $\tau^{asset} \approx 0.002$ and $\tau^{curr} \approx 0.0002$.

Bibliography

[1] Manuel Blum, Robert W Floyd, Vaughan R Pratt, Ronald L Rivest, and Robert E Tarjan. Time bounds for selection. *Journal of Computer and System Sciences*, 7(4):448–461, 1973.

[2] Raphael Burkhardt Urban Ulrych. Sparse and stable international portfolio optimization and currency risk management, 2023.

Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Declaration of originality

The signed declaration of originality is a component of every written paper or thesis authored during the course of studies. **In consultation with the supervisor**, one of the following two options must be selected:

☐ I hereby declare that I authored the work in question independently, i.e. that no one helped me to author it. Suggestions from the supervisor regarding language and content are excepted. I used no generative artificial intelligence technologies[1].

☐ I hereby declare that I authored the work in question independently. In doing so I only used the authorised aids, which included suggestions from the supervisor regarding language and content and generative artificial intelligence technologies. The use of the latter and the respective source declarations proceeded in consultation with the supervisor.

Title of paper or thesis:

Authored by:
If the work was compiled in a group, the names of all authors are required.

Last name(s):	**First name(s):**

With my signature I confirm the following:
- I have adhered to the rules set out in the Citation Guidelines.
- I have documented all methods, data and processes truthfully and fully.
- I have mentioned all persons who were significant facilitators of the work.

I am aware that the work may be screened electronically for originality.

Place, date	**Signature(s)**

If the work was compiled in a group, the names of all authors are required. Through their signatures they vouch jointly for the entire content of the written work.

[1] For further information please consult the ETH Zurich websites, e.g. https://ethz.ch/en/the-eth-zurich/education/ai-in-education.html and https://library.ethz.ch/en/researching-and-publishing/scientific-writing-at-eth-zurich.html (subject to change).